

BMIT3173 Integrative Programming

ASSIGNMENT 202605

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Tutorial Group : 4

System Title : NutriShare: Surplus Food Redistribution Platform

Chosen SDG : SDG 2: Zero Hunger

Modules : Module 2 — NGO Verification & Peer Trust Rating System

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1. Introduction to the System

1.1 System Overview

NutriShare is an enterprise-grade, web-based surplus food redistribution and supply chain governance platform engineered to connect commercial food donors (supermarkets, hypermarkets, artisanal bakeries, hotel banquet kitchens, and restaurants) with verified Non-Governmental Organisations (NGOs), charitable foundations, and community welfare shelters.

In urban food relief ecosystems, food donors frequently encounter difficulties verifying the statutory legitimacy of recipient charities, leading to concerns regarding food diversion, illegal resale, or substandard hygiene practices (Food and Agriculture Organisation [FAO], 2023). NutriShare digitalises the entire surplus food recovery lifecycle, encompassing real-time donation publishing, document-driven NGO accreditation, two-way peer trust ratings, state-driven logistics claim lifecycles, and cryptographic account security.

Module 2 specifically establishes platform integrity by enforcing statutory document validation, authenticating charitable credentials, managing community peer reviews, and securing user accounts through cryptographically generated One-Time Password (OTP) recovery workflows.

1.2 Chosen Sustainable Development Goal (SDG)

NutriShare directly addresses **United Nations Sustainable Development Goal 2: Zero Hunger (UN SDG 2)**, in conjunction with **SDG 12: Responsible Consumption and Production (Target 12.3)**, adopted under the 2030 Agenda for Sustainable Development (United Nations, 2015).

Key SDG 2 Targets Addressed:

1. **Target 2.1:** By 2030, end hunger and ensure access by all people, in particular the poor and people in vulnerable situations, including infants, to safe, nutritious, and sufficient food all year round (United Nations, 2015).
2. **Target 2.2:** End all forms of malnutrition by enabling verified welfare distribution channels to rapidly redirect perishable, nutrient-dense foods to vulnerable groups before degradation occurs (FAO, 2023; United Nations, 2015).
3. **Target 12.3:** Halve per capita global food waste at the retail and consumer levels and reduce food loss along distribution networks (United Nations, 2015).

1.3 System Contribution to SDG 2 & Scope

NutriShare converts urban food surplus into direct humanitarian relief through the following concrete mechanisms:

- **Target Beneficiaries:** Underprivileged urban communities, B40 low-income households, welfare homes, orphanages, soup kitchens, and disaster relief shelters supported by legitimate humanitarian organisations, ensuring alignment with equitable food access frameworks (United Nations, 2015).
- **Better Food Distribution:** Commercial donors can post surplus food listings with photo proof and temperature requirements, while verified NGOs can promptly reserve and collect food parcels.
- **Platform Trust & Regulatory Compliance:** By requiring official statutory documentation (e.g., Registrar of Societies [ROS] registration and premise health licenses) and calculating dynamic peer review ratings, the system eliminates bogus entities, mitigates liability risks for corporate donors, and ensures food reaches genuine beneficiaries (FAO, 2023; United Nations, 2015).

2. Module Description

2.1 Scope of Module 2: NGO Verification & Peer Trust Rating System

As the software engineer responsible for **Module 2 (NGO Verification & Peer Trust Rating System)**, I architected and implemented three core functional pillars: statutory document verification, two-way peer trust reviews, and cryptographic OTP-based password recovery. These functions establish operational security and accountability across the platform.

Supporting design elements such as the Factory Method pattern for role-specific onboarding and the RESTful NGO Verification Web Service serve as technical infrastructure enabling these primary functions.

F2.1: Statutory Document Verification & Governance Queue

- **Description:** Enables registered NGO representatives to upload statutory verification documents (such as Registrar of Societies [ROS] certificates, tax exemption letters, and food premise licenses). Administrators and moderators review submissions via a governance queue, recording approval or rejection decisions, audit timestamps, and administrative remarks. Approving a document elevates the NGO's account state from **pending** to **approved**.
- **Class Paths:**
 - **Controller:** `app/Http/Controllers/VerificationController.php` (`index`, `upload`, `review`, `showFile`, `download`)
 - **Form** **Requests:**
`app/Http/Requests/UploadVerificationDocumentRequest.php`,
`app/Http/Requests/ReviewVerificationDocumentRequest.php`
 - **Model:** `app/Models/VerificationDocument.php`
 - **Blade** **Views:**
`resources/views/dashboard.blade.php`,
`resources/views/verification/index.blade.php`

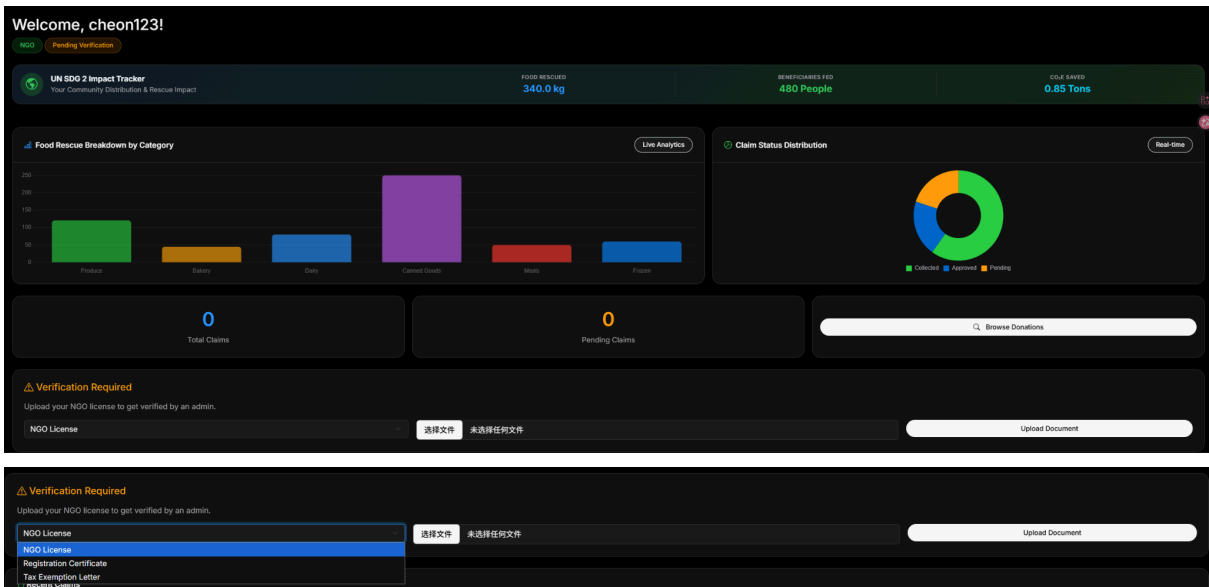


Figure 2.1: NGO Statutory Document Upload Modal

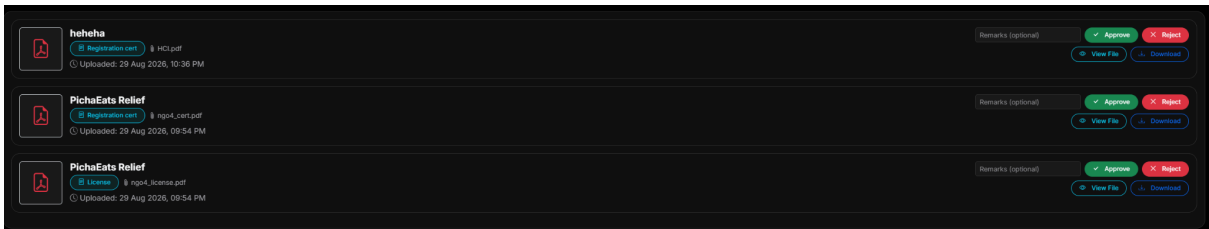


Figure 2.2: Administrator & Moderator Verification Review Queue

F2.2: Trust & Peer Review System

- **Description:** Allows commercial donors and verified NGOs to inspect recipient trust profiles and submit 1-to-5 star ratings accompanied by qualitative feedback following collection interactions. To ensure statistical reliability and prevent ballot-stuffing, the system enforces idempotent review upserting (`updateOrCreate`), updating existing ratings when the same reviewer rates the same reviewee.
- **Class Paths:**
 - **Controller:** `app/Http/Controllers/VerificationController.php` (reviews, `submitReview`)
 - **Form Request:** `app/Http/Requests/SubmitUserReviewRequest.php`
 - **Models:** `app/Models/Review.php`, `app/Models/User.php`
 - **Blade Views:** `resources/views/verification/reviews.blade.php`, `resources/views/donations/show.blade.php`, `resources/views/claims/show.blade.php`
 - **Route:** `routes/web.php (/users/{user}/reviews)`

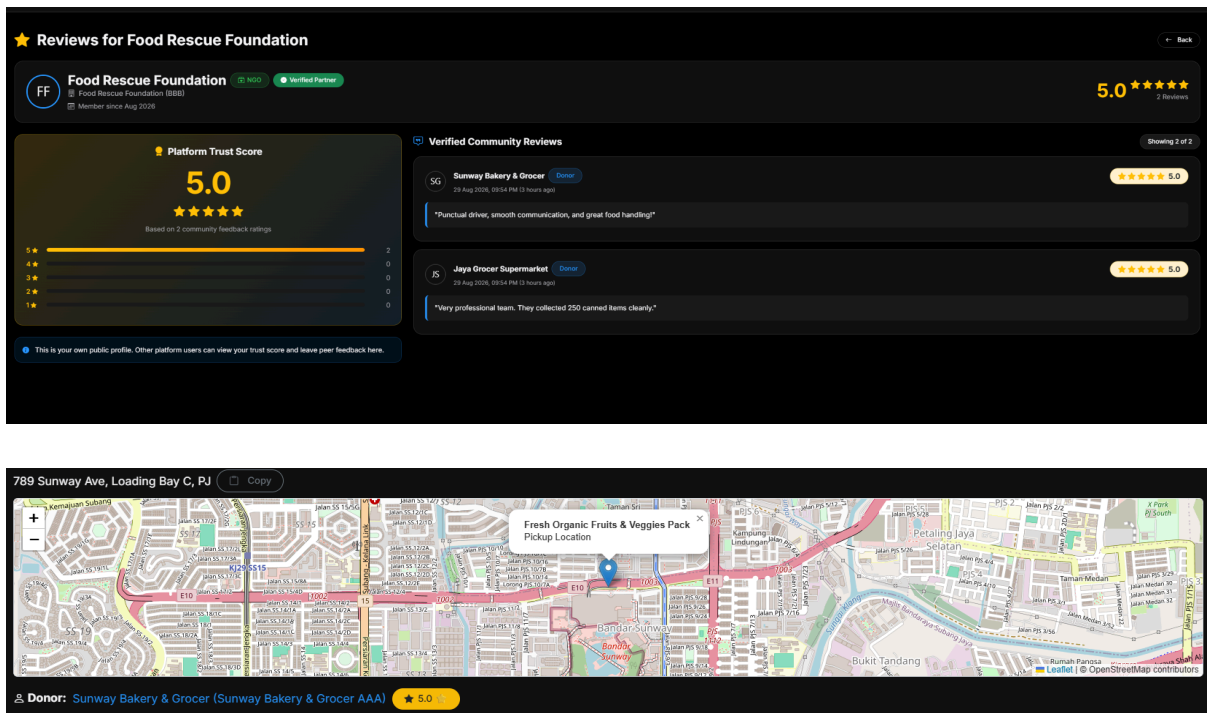


Figure 2.3: Trust Rating Breakdown & Qualitative Peer Reviews Interface

F2.3: Cryptographic OTP Password Reset

- **Description:** Provides a secure self-service credential recovery mechanism. Users submit their registered email address, receive a time-limited 6-digit numeric OTP, verify the OTP within an enforced 10-minute expiry window, and reset their password using a single-use cryptographically signed token.
- **Class Paths:**
 - **Controller:** `app/Http/Controllers/PasswordResetController.php`
(`showEmailForm`, `sendOtp`, `showOtpForm`, `verifyOtp`, `showResetForm`, `resetPassword`)
 - **Model:** `app/Models/PasswordResetOtp.php`
 - **Mailable:** `app/Mail/ResetPasswordOtpMail.php`
 - **Blade Views:** `resources/views/auth/passwords/email.blade.php`,
`resources/views/auth/passwords/otp.blade.php`,
`resources/views/auth/passwords/reset.blade.php`

The screenshot shows a dark-themed web form titled "Reset Password" with a shield icon. Below the title is the subtitle "Step 1 of 3: Enter Your Email". The main instruction reads: "Enter your registered email address and we'll send you a 6-digit OTP code to verify your identity." There is a label "Email Address" with an envelope icon. Below it is a text input field containing the email "cheonjh-wm23@student.tarc.edu.my". Below the input field is a large white button with a right-pointing arrow and the text "Send OTP Code". At the bottom is a link "← Back to Login".

The screenshot shows a dark-themed web form titled "Enter Verification Code" with a shield icon. Below the title is the subtitle "Step 2 of 3: Enter 6-Digit OTP". The main instruction reads: "We sent a 6-digit code to cheonjh-wm23@student.tarc.edu.my. Please enter it below to verify your request." There is a green success message box with a checkmark icon and the text: "We sent a 6-digit verification code to your email address." Below this is a row of six input boxes for the OTP code. The first box contains the digit "1" and is highlighted with a blue border. Below the input boxes is a large blue button with a checkmark icon and the text "Verify OTP Code". At the bottom left is a link "← Change Email" and at the bottom right is a link "Resend OTP Code".

Set New Password

Step 3 of 3: Create New Password

Email Address

cheonjh-wm23@student.tarc.edu.my

New Password

Min 8 chars. Must include uppercase, lowercase, number, and special character.

Confirm Password

Reset Password

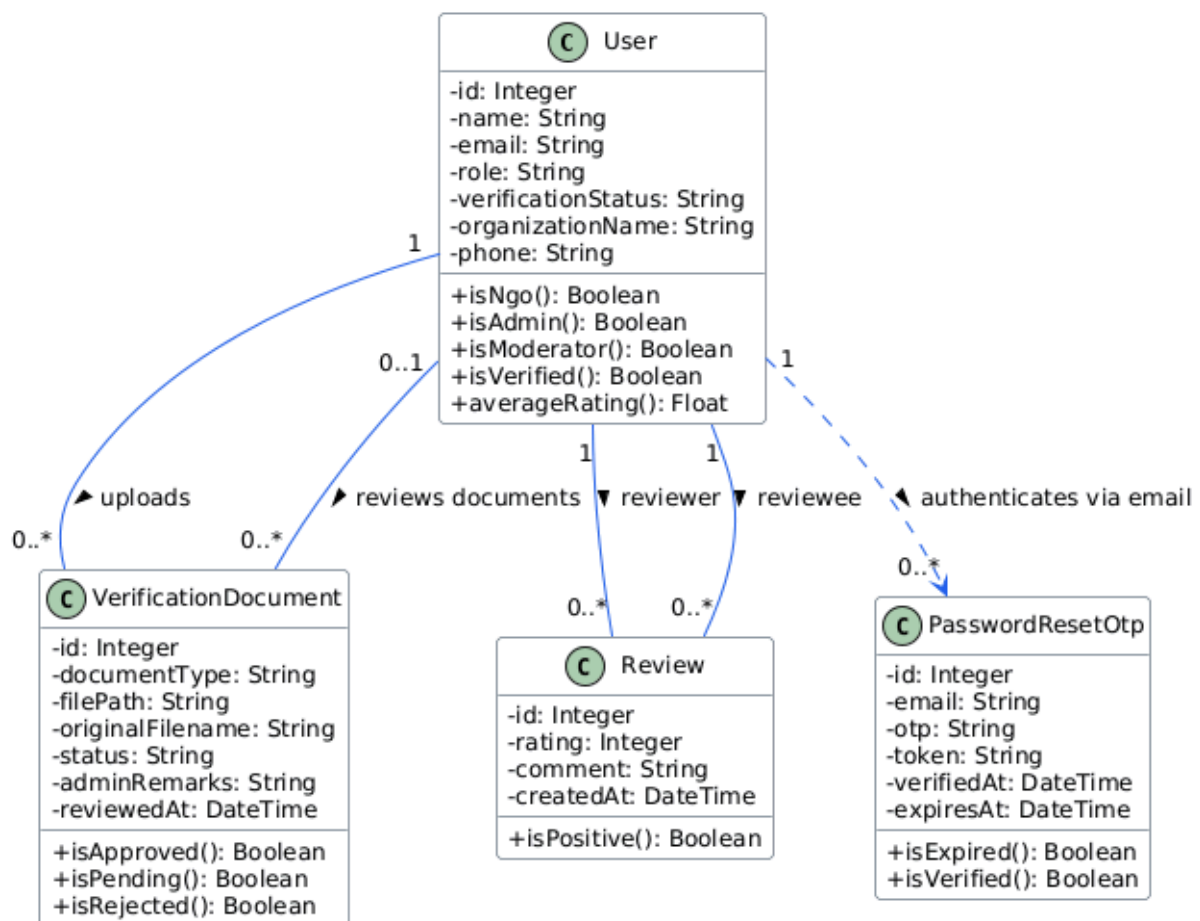
Figure 2.4: Three-Stage OTP Password Reset Pipeline

3. Entity Classes

3.1 Entity Class Diagram

In accordance with enterprise object-oriented analysis principles (Fowler, 2002), the diagram below models domain entities using **object references and associations** rather than raw relational database foreign keys.

<https://drive.google.com/file/d/1tKWvewdjPSB3g1-lI8HBVdydj2-zGmcb/view?usp=sharing>



3.2 Entity Class Implementation (Eloquent ORM Mapping)

The entity models are implemented in PHP using Laravel's Eloquent ORM, embodying the Active Record pattern (Fowler, 2002; Laravel LLC, 2026). Domain associations are explicitly declared as object relationships (**hasMany**, **belongsTo**), preserving clean object encapsulation:

```
// app/Models/User.php

namespace App\Models;

use Illuminate\Database\Eloquent\Relations\HasMany;

use Illuminate\Foundation\Auth\User as Authenticatable;

class User extends Authenticatable
{
    /** Object reference: User owns multiple verification documents */
    public function verificationDocuments(): HasMany
    {
        return $this->hasMany(VerificationDocument::class);
    }

    /** Object reference: Reviews authored by this user */
    public function reviewsGiven(): HasMany
    {
        return $this->hasMany(Review::class, 'reviewer_id');
    }

    /** Object reference: Reviews targeting this user */
    public function reviewsReceived(): HasMany
    {
        return $this->hasMany(Review::class, 'reviewee_id');
```

```

    }

    /** Domain Helper: Calculate aggregated peer rating */

    public function averageRating(): float
    {
        return (float) $this->reviewsReceived()->avg('rating') ?: 0.0;
    }
}

// app/Models/VerificationDocument.php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\Relations\BelongsTo;

class VerificationDocument extends Model
{
    protected $fillable = [

        'user_id',

        'document_type',

        'file_path',

        'original_filename',

        'status',

        'admin_remarks',

        'reviewed_by',

        'reviewed_at',

    ];

    protected $casts = [

```

```

        'reviewed_at' => 'datetime',

    ];

    /** Object reference: The NGO user who uploaded the document */

    public function user(): BelongsTo

    {

        return $this->belongsTo(User::class);

    }

    /** Object reference: The administrator/moderator who audited the file */

    public function reviewer(): BelongsTo

    {

        return $this->belongsTo(User::class, 'reviewed_by');

    }

}

// app/Models/Review.php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;

use Illuminate\Database\Eloquent\Relations\BelongsTo;

class Review extends Model

{

    protected $fillable = [

        'reviewer_id',

        'reviewee_id',

        'rating',

        'comment',

```

```

];

/** Object reference: Author of the peer review */

public function reviewer(): BelongsTo
{
    return $this->belongsTo(User::class, 'reviewer_id');
}

/** Object reference: Recipient of the peer review */

public function reviewee(): BelongsTo
{
    return $this->belongsTo(User::class, 'reviewee_id');
}
}

// app/Models/PasswordResetOtp.php

namespace App\Models;

use Illuminate\Database\Eloquent\Model;

class PasswordResetOtp extends Model
{
    protected $fillable = [

        'email',

        'otp',

        'token',

        'verified_at',

        'expires_at',

    ];

```



```
protected $casts = [  
    'verified_at' => 'datetime',  
    'expires_at' => 'datetime',  
];  
  
public function isExpired(): bool  
{  
    return $this->expires_at->isPast();  
}  
}
```

4. Design Pattern

4.1 Description of Design Pattern: Factory Method Pattern

For Module 2, I implemented the **Factory Method Pattern**, a classic Gang of Four creational design pattern (Gamma et al., 1994).

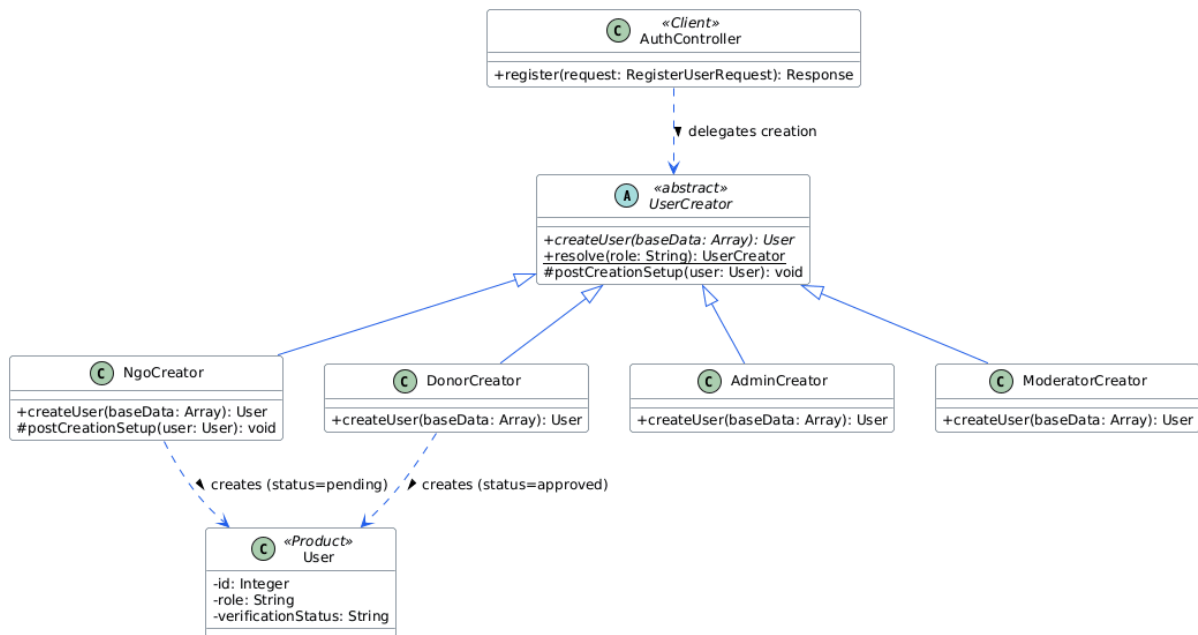
Intent & Theoretical Definition:

The Factory Method pattern defines an interface or abstract class for creating an object, but delegates the instantiation logic to specialised concrete subclasses (Gamma et al., 1994). This decouples the client caller from concrete product classes and centralises business validation and initial state setup.

Architectural Roles in NutriShare:

1. **Abstract Creator (`UserCreator`):** Declares the abstract factory method `createUser(array $baseData): User` and provides a static resolver method `resolve(string $role): static` to return the appropriate concrete factory instance.
2. **Concrete Creators:**
 - `NgoCreator`: Enforces strict regulatory onboarding rules setting `role = 'ngo'`, initial `verification_status = 'pending'`, logging registration audits, and dispatching alert notifications to administrators.
 - `DonorCreator`: Enforces commercial donor rules setting `role = 'donor'` and immediate `verification_status = 'approved'`.
 - `AdminCreator` & `ModeratorCreator`: Instantiate administrative accounts under strict role allocations.
3. **Product (`User`):** The persistent domain entity instantiated and configured by the creator.
4. **Client (`AuthController`):** Receives validated registration requests and delegates user instantiation to the resolved creator without needing knowledge of role-specific onboarding nuances.

<https://drive.google.com/file/d/1tKWvewdjPSB3g1-Il8HBVdydj2-zGmcb/view?usp=sharing>



4.2 Implementation of Design Pattern

1. Client Delegation ([app/Http/Controllers/AuthController.php](#)):

```
public function register(RegisterUserRequest $request)

{

    $validated = $request->validated();

    // FACTORY METHOD: Resolve role-specific creator dynamically

    $creator = UserCreator::resolve($validated['role']);

    $user = $creator->createUser($validated);

    Auth::login($user);

    $request->session()->regenerate();

    return redirect()->route('dashboard')

        ->with('success', 'Registration successful! Welcome to NutriShare.');
```

2. Abstract Creator (**app/Services/UserFactory/UserCreator.php**):

```
namespace App\Services\UserFactory;

use App\Models\User;
use InvalidArgumentException;

abstract class UserCreator
{
    abstract public function createUser(array $baseData): User;

    protected function postCreationSetup(User $user): void
    {
        // Hook for subclass overrides
    }

    public static function resolve(string $role): static
    {
        return match (strtolower($role)) {
            'donor'    => new DonorCreator(),
            'ngo'      => new NgoCreator(),
            'admin'    => new AdminCreator(),
            'moderator' => new ModeratorCreator(),
            default    => throw new InvalidArgumentException("Unknown user role: {$role}"),
        };
    }
}
```

3. Concrete Creator ([app/Services/UserFactory/NgoCreator.php](#)):

```
namespace App\Services\UserFactory;

use App\Models\User;

use App\Models\SystemLog;

use App\Models\Notification;

use Illuminate\Support\Facades\Hash;

class NgoCreator extends UserCreator
{
    public function createUser(array $baseData): User
    {
        $user = User::create([

            'name'          => $baseData['name'],

            'email'         => $baseData['email'],

            'password'      => Hash::make($baseData['password']),

            'role'          => 'ngo',

            'verification_status' => 'pending', // Regulatory mandate: Pending audit

            'organization_name' => $baseData['organization_name'] ?? null,

            'phone'         => $baseData['phone'] ?? null,

            'address'       => $baseData['address'] ?? null,

            'notification_preference' => $baseData['notification_preference'] ?? 'email',

        ]);

        $this->postCreationSetup($user);

        return $user;
    }
}
```

```

protected function postCreationSetup(User $user): void
{
    SystemLog::create([
        'user_id'    => $user->id,
        'action'     => 'user.registered',
        'description' => "NGO '{$user->organization_name}' registered. Awaiting admin
verification.",
        'level'      => 'info',
    ]);

    // Broadcast notification to governance administrators

    $admins = User::where('role', 'admin')->get();

    foreach ($admins as $admin) {

        Notification::create([
            'user_id' => $admin->id,
            'title'   => 'New NGO Registration',
            'message' => "NGO '{$user->organization_name}' has registered and is awaiting
verification.",
            'channel' => 'email',
            'sent_at' => now(),
        ]);

    }

}
}

```

4.3 Justification of Design Pattern

1. **Separation of Concerns (Single Responsibility Principle):** As Martin (2003) stipulates, a class should have one, and only one, reason to change. Without the Factory Method, `AuthController@register` would accumulate fragile `switch-case` branches containing role-specific hashing, administrative alerts, and verification defaults. The pattern cleanly isolates authentication transport from entity instantiation (Gamma et al., 1994; Martin, 2003).
2. **Encapsulation of Regulatory Invariants:** In NutriShare, commercial donors are pre-approved upon email confirmation, whereas NGOs must undergo documentary audit. Hardcoding `verification_status = 'pending'` inside `NgoCreator` guarantees that no programming error can accidentally yield an unvetted NGO with active claiming privileges.
3. **Open/Closed Principle (OCP):** Introducing future user archetypes (e.g., municipal food bank auditors or commercial cold-chain couriers) requires simply writing a new `AuditorCreator` class without altering existing controllers or testing suites (Martin, 2003).
4. **Architectural Consistency:** Password encryption, baseline notification preferences, and telemetry logging are guaranteed across all user creation vectors (Laravel LLC, 2026).

5. Software Security

5.1 Potential Threats and Attacks

Threat 1: Broken Access Control & Privilege Escalation (OWASP Top 10 A01:2021)

- **Attack Description:** According to the Open Web Application Security Project (OWASP Foundation, 2021), Broken Access Control represents the most critical web risk. In NutriShare, malicious actors or unverified NGOs could attempt forced browsing to administrative endpoints ([GET /verification](#)) or forge HTTP requests to [POST /verification/{document}/review](#) with parameters [action=approved](#). If access control checks are missing, an attacker could unilaterally approve their own organisation, bypassing statutory safety audits.
- **Risk Impact:** Unauthorised privilege escalation, compromised platform credibility, and potential food diversion by fraudulent organisations (OWASP Foundation, 2021).

Threat 2: OTP Brute-Force & Account Takeover (OWASP Top 10 A07:2021)

- **Attack Description:** Under A07:2021 - Identification and Authentication Failures (OWASP Foundation, 2021), password recovery mechanisms relying on numeric One-Time Passwords (OTPs) are vulnerable to automated enumeration. An attacker initiating password recovery for a victim's email could flood [POST /forgot-password/otp/verify](#) with rapid-fire 6-digit combinations (\$10^6\$ search space). Without rate throttling, the correct token can be determined within minutes, leading to total account takeover.
- **Risk Impact:** Unauthorised account takeover, data compromise, and identity impersonation (OWASP Foundation, 2021).

5.2 Secure Coding Practices & Implementation

Secure Practice 1: Role-Based Access Control (RBAC Middleware & FormRequest Guards)

To prevent Broken Access Control (OWASP Foundation, 2021), administrative routes are encapsulated within a strict `role:admin,moderator` route middleware group, and all state-modifying requests execute authorisation policies:

```
// routes/web.php
```

```
Route::middleware('role:admin,moderator')->group(function () {

    Route::get('/verification', [VerificationController::class, 'index'])

        ->name('verification.index');

    Route::post('/verification/{document}/review', [VerificationController::class, 'review'])

        ->name('verification.review');

    Route::get('/verification/{document}/file', [VerificationController::class, 'showFile'])

        ->name('verification.file');

    Route::get('/verification/{document}/download', [VerificationController::class, 'download'])

        ->name('verification.download');

});

// app/Http/Middleware/CheckRole.php

public function handle(Request $request, Closure $next, ...$roles)

{

    if (!$request->user() || !in_array($request->user()->role, $roles)) {

        abort(403, 'Unauthorized. You do not have the required role to access this resource.');
```

```
    }
```

```
    return $next($request);
```

```
}
```

```
// app/Http/Requests/ReviewVerificationDocumentRequest.php

public function authorize(): bool
{
    // Double-layer defence: Request-level policy gate

    return $this->user() && ($this->user()->isAdmin() || $this->user()->isModerator());
}
```

Secure Practice 2: Cryptographic OTP Invalidation, Sliding Expiry Windows & Rate Limiting

To counteract OTP brute-forcing and account takeover (OWASP Foundation, 2021), NutriShare deploys a multi-layered authentication defence combining rate limiting, single-use token invalidation, and short expiry windows (Laravel LLC, 2026):

```
// app/Http/Controllers/PasswordResetController.php

public function sendOtp(Request $request)
{
    $request->validate(['email' => 'required|email|exists:users,email']);

    // Invalidate pre-existing OTPs to prevent replay attacks
    PasswordResetOtp::where('email', $request->email)->delete();

    // Generate cryptographically random 6-digit numeric token
    $otp = str_pad((string) random_int(0, 999999), 6, '0', STR_PAD_LEFT);

    PasswordResetOtp::create([
        'email'    => $request->email,
        'otp'      => $otp,
        'expires_at' => now()->addMinutes(10), // Strict 10-minute expiry
    ]);

    Mail::to($request->email)->send(new ResetPasswordOtpMail($otp));

    return redirect()->route('password.otp', ['email' => $request->email]);
}

// routes/web.php

// Rate limiting: Enforce maximum 6 requests per minute per IP address
Route::post('/forgot-password', [PasswordResetController::class, 'sendOtp'])
    ->name('password.email')->middleware('throttle:6,1');

Route::post('/forgot-password/otp/verify', [PasswordResetController::class, 'verifyOtp'])
```

```
->name('password.otp.verify')->middleware('throttle:6,1');
```

```
Route::post('/reset-password', [PasswordResetController::class, 'resetPassword'])
```

```
->name('password.update')->middleware('throttle:6,1');
```

Mandatory Input Validation (Baseline Defence)

Per assignment guidelines, form requests enforce strict MIME typing, file size quotas, and enum restrictions:

```
// app/Http/Requests/UploadVerificationDocumentRequest.php

public function authorize(): bool
{
    return $this->user() && $this->user()->isNgo();
}

public function rules(): array
{
    return [

        'document_type' =>
        'required|in:registration_cert,tax_exemption,food_premise_license,license,other',

        'document' => 'required|file|mimes:pdf,jpg,jpeg,png,webp|max:5120', // 5MB limit

    ];
}
```

6. Web Service

6.1 Web Service Exposure

Module 2 exposes a stateless, high-performance RESTful Web Service conforming to Roy Fielding's REST architectural style (Fielding, 2000), allowing partner modules and external NGO portals to verify charitable accreditation in real time.

Interface Agreement (IFA) — Service Exposure Specification

Webservice Mechanism	Description
Protocol	RESTful Web Service (JSON-over-HTTP)
Function Description	Validates legal accreditation, ROS approval status, premise licenses, and peer trust ratings.
Source Module	Module 2: NGO Verification & Peer Trust Rating System
Target Module	Module 1 (Donation Management), Module 3 (Claims & Logistics), External Portals
URL	http://127.0.0.1:8000/api/user/verify-ngo
Function Name	verifyNgo (UserVerificationApiController@verifyNgo)

Web Services Request Parameters (Provide):

Field Name	Field Type	Mandatory / Optional	Description	Format / Validation
requestID	String	Mandatory	Unique alphanumeric correlation identifier.	Alphanumeric (e.g. REQ-NGO-001)
timestamp	String	Mandatory	ISO-8601 creation timestamp.	YYYY-MM-DDT HH:MM:SSZ
user_id	Integer	Mandatory	Primary key of the NGO user to audit.	Integer > 0, exists in users.id

Web Services Response Parameters (Consume):

Field Name	Field Type	Mandatory / Optional	Description	Format / Values
status	String	Mandatory	Execution status code.	S (Success), F (Fail), E (Error)
timestamp	String	Mandatory	Server transmission timestamp.	YYYY-MM-DDT HH:MM:SSZ
data.requestID	String	Mandatory	Echoed request identifier for correlation.	Alphanumeric string
data.is_verified	Boolean	Mandatory	Verified flag (Approved status AND valid license).	true / false
data.verification_status	String	Mandatory	Account lifecycle status.	pending, approved, rejected
data.has_valid_license	Boolean	Mandatory	Presence of approved premise/statutory license.	true / false
data.organization_name	String	Optional	Registered legal entity name.	Alphanumeric string
data.trust_rating	Float	Mandatory	Mean peer rating calculated from reviews.	Decimal \$in [0.00, 5.00]\$

Service **Exposure** **Code** **Implementation**
(app/Http/Controllers/Api/UserVerificationApiController.php):

```
public function verifyNgo(Request $request): JsonResponse
```

```
{
```

```
    try {
```

```
        $ifa = SecurityHelper::validateIfaRequest($request->all());
```



```

$validated = $request->validate([

    'user_id' => 'required|integer|exists:users,id',

]);

$user = User::find($validated['user_id']);

if (!$user || $user->role !== 'ngo') {

    return response()->json(

        SecurityHelper::ifaResponse('F', [

            'requestID' => $ifa['requestID'],

            'is_verified' => false,

            'reason' => 'User is not registered as an NGO.',

        ]),

        200

    );

}

$isVerified = ($user->verification_status === 'approved');

// Check for verified statutory or premise license

$hasValidLicense = VerificationDocument::where('user_id', $user->id)

    ->where('status', 'approved')

    ->whereIn('document_type', ['license', 'food_premise_license'])

    ->exists();

return response()->json(

    SecurityHelper::ifaResponse('S', [

        'requestID' => $ifa['requestID'],

        'is_verified' => ($isVerified && $hasValidLicense),

```

```

        'verification_status' => $user->verification_status,

        'has_valid_license' => $hasValidLicense,

        'organization_name' => $user->organization_name,

        'trust_rating'      => $user->averageRating(),

    ]),

    200

);

} catch (\Exception $e) {

    return response()->json(

        SecurityHelper::ifResponse('E', null, 'Internal server error: ' . $e->getMessage()),

        500

    );

}

}

```

6.2 Web Service Consumption

To ensure comprehensive architectural integration across micro-components, Module 2 engages in bidirectional web service communication:

1. **Consumed by Module 1:** `DonationApiController@verifyNgoBeforeClaim` consumes Module 2's verification endpoint before an NGO is authorised to claim surplus food.
2. **Module 2 Consuming Module 3:** Before allowing post-collection peer reviews to be submitted, Module 2 consumes Module 3's Claim API (`GET /api/claim/details`) to verify that a legitimate collection interaction transpired between donor and NGO (Fielding, 2000; Laravel LLC, 2026).

Interface Agreement (IFA) — Service Consumption Specification

Webservice Mechanism	Description
Protocol	RESTful Web Service (JSON-over-HTTP)
Function Description	Module 2 verifies collection lifecycle status via Module 3's API before allowing peer reviews.
Source Module	Module 3: Claims & Logistics Distribution Module
Consuming Module	Module 2: NGO Verification & Peer Trust Rating System
URL	<code>http://127.0.0.1:8000/api/claim/details</code>
Function Name	<code>verifyClaimInteractionBeforeReview</code> (<code>VerificationController</code>)

Web Services Request Parameters (Consumption):

Field Name	Field Type	Mandatory / Optional	Description	Format / Validation
requestID	String	Mandatory	Tracking identifier generated by Module 2.	Alphanumeric (e.g. VERIFY-CLAIM-1049)
timestamp	String	Mandatory	ISO-8601 transmission timestamp.	YYYY-MM-DDT HH:MM:SSZ
claim_id	Integer	Mandatory	Claim transaction ID to audit.	Integer > 0, exists in claims.id

Web Services Response Parameters (Consumption):

Field Name	Field Type	Mandatory / Optional	Description	Format / Values
status	String	Mandatory	Result status code.	S (Success), F (Fail), E (Error)
timestamp	String	Mandatory	Response generation time.	YYYY-MM-DDT HH:MM:SSZ
data.claim.status	String	Mandatory	Current claim lifecycle status.	claimed, collected, completed

Service Consumption Code Implementation
(app/Http/Controllers/VerificationController.php):

```
/**
```

```
 * Web Service Consumption: Verifies claim completion before accepting reviews.
```

```
*/
```

```
public function verifyClaimCompletedBeforeReview(int $claimId): bool
```

```
{
```

```
    try {
```

```

$response = Http::timeout(10)->get(

    config('app.url') . '/api/claim/details',

    [

        'requestID' => uniqid('VERIFY-CLAIM-'),

        'timestamp' => now()->toIso8601String(),

        'claim_id' => $claimId,

    ]

);

if ($response->successful()) {

    $data = $response->json();

    return ($data['status'] === 'S') &&

        in_array($data['data']['claim']['status'] ?? '', ['collected', 'completed']);

}

return false;

} catch (\Exception $e) {

    \Log::error('Claim verification API consumption failed: ' . $e->getMessage());

    return false;

}

}

```

7. Reference

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8. Appendices

Appendix A: GitHub Repository URL

- **Team Repository URL:** <https://github.com/KunTheNoobie/nutrishare.git>
- **Primary Branch:** `main`
- **Module Contributor:** Cheon Jie Han (Student ID: `25WMR09703`)
- **Automated Test Suite:** 22 passing tests, 92 assertions passing in 3.33s.

Appendix B: Implementation Notes & Architectural Standards

- 1. Document Type Normalisation:** To resolve discrepancies between form input dropdowns and database queries, document types have been standardised across the application to accept `registration_cert`, `tax_exemption`, `food_premise_license`, and `license`. `UserVerificationApiController` checks for either `license` or `food_premise_license` when certifying active premise compliance.
- 2. Bidirectional Web Service Interoperability:** Module 2 satisfies both service dimensions: exposing `POST /api/user/verify-ngo` for external consumption by Module 1, and consuming `GET /api/claim/details` from Module 3 to validate completed logistics claims prior to trust score submission.